The Spectre Knot

One tile, front and back

Front: the knot tile. Back: what it is. Edge length 41 mm.



The Spectre Knot

A single-strand Celtic knot on an aperiodic monotile

The same ribbon on every tile of a tiling that never repeats, and two facts about it. In every spectre tiling that grows out in ever-larger blocks to fill the plane, no strand ever closes into a loop. And in the tiling grown outwards from one tile, each block sitting in the same place inside the next, there is just **one strand:** every piece joins one endless Celtic knot.

The tile: the straight-edged spectre, discovered in 2023 by David Smith, Joseph Samuel Myers, Craig S. Kaplan and Chaim Goodman-Strauss. **The knot:** Justin Roughley came up with the original idea of looking for a mono-stranded aperiodic knot, and then used AI to search for it (2026). Made with AI: cortado-j.github.io/SPECTRE_KNOT

Justin Roughley,
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1. Cut round the outer black line, as one piece.
2. Fold along the dashed line, printed sides out.
3. Glue the halves back to back.

Make several and lay them edge to edge: the ribbons meet where the edges touch and cross there at right angles.